Tier 1 | Vectors to a working store

The core contract

A Record has an ID, vector, metadata and optional text. A CollectionSpec fixes dimension, metric and model space. Validate finite values and reject zero vectors for cosine search. Equal dimensions do not imply equal model spaces.

Three formulas

dot(x,q) = sum(x_i*q_i)
cosine(x,q) = dot(x,q)/(norm(x)*norm(q))
L2_squared(x,q) = sum((x_i-q_i)^2)
float32 vector bytes = rows * dimensions * 4

Build TinyVec

Validate the complete batch, replace existing IDs, invalidate the derived index, then search only eligible records. Persist schema, vectors and metadata together; restore in a fresh process and compare IDs. Never load untrusted arbitrary-object serialization.

Measure honestly

ANN recall compares retrieved neighbors with exact neighbors. Relevance uses human or synthetic judgments. State k, tie handling, filter eligibility and dtype. Time embedding, build and retrieval separately; report sample counts with percentiles.

Run from the repository root

uv run python examples/01_quickstart.py
uv run python examples/02_filtered_search.py
uv run python scripts/checkpoint_tier1.py

Before moving on

Explain why the mug-warmer query works without the exact phrase. Show replacement, deletion and restore. Confirm model revision/preprocessing in the vector manifest. Included real-model vectors let the first search run without downloading a model. See Modules 00-04 and docs/MENTAL_MODEL.md.